

(Please write your Exam Roll No.)

Exam Roll No.

END TERM EXAMINATION

FIFTH SEMESTER [B.TECH] DECEMBER- 2018

Paper Code: ETEC-305

Subject: Microprocessors & Microcontrollers

Time: 3 Hours

Maximum Marks: 75

Note: Attempt any five questions including Q.No1 which is compulsory.
Attempt one question from each unit.

- Q1 (a) What determines whether a microprocessor is considered a 4-bit, a 16-bit and 32-bit device?
(b) Explain intra segment and inter segment jump.
(c) Explain the function of SIM and RIM instruction.
(d) Explain the assembler directives: EQU and DB used in 8051.
(e) Explain the flag register of microprocessor 8085 and 8086.
(f) What is the use of BSR mode in 8255?
(g) Write a program to toggle P3.2 using Timer interrupt in 8051.
(h) Explain the function of TMOD register in 8051.
(i) Write a program to toggle pin P1.2 every second.
(j) Explain PUSH and POP instruction in 8051. (2.5x10=25)

UNIT-I

- Q2 (a) Draw and explain the architecture of 8085. (6)
(b) Draw and explain timing diagram for the instruction MVI B, 05h. (6.5)
- Q3 (a) Write a program in 8085 to:- (3x2=6)
(i) Find largest number in a given string.
(ii) Find even number in a given string.
(b) Write a program in 8085 to generate Fibonacci series. (6.5)

UNIT-II

- Q4 (a) Explain minimum mode and maximum mode configuration of 8086. (6)
(b) Explain various interrupts of 8086 in detail and explain how to calculate vectored address of interrupts. (6.5)
- Q5 (a) Interface two 4Kx8 EPROM chips and two 4Kx8 RAM chips with 8086. Select suitable memory maps. (6.5)
(b) Explain function of instruction with suitable examples. (6)
(i) LEA (ii) DAA (iii) LOOP (iv) LDS (v) IN (vi) PUSH

UNIT-III

- Q6 (a) Draw the block diagram of 8255 and explain its interfacing with 8086. (6)
(b) Draw the block diagram of 8253 and explain its different modes of operation. (6.5)
- Q7 (a) Interface 8 bit Analog to Digital converter with microprocessor using 8255 IC. Draw the block diagram of interfacing; decode hex address for the ports and write an ALP for its functioning. (10)
(b) Explain asynchronous mode of USART 8251? (2.5)

UNIT-IV

- Q8 (a) Explain the 8051 microcontroller block diagram in detail. (6)
(b) Write an ALP to check if the character string of length 7, stored in RAM locations 40H onwards is a palindrome output 'Y' to P1. (6.5)
- Q9 (a) Write a program to implement interrupt INT0, INT1. Glow all LEDs of P0 when INT0 is activated & glow all LEDs of P2 when INT1 is activated. (6)
(b) Write a program to generate two square waves of frequency 5KHz at pin p1.3 and another of frequency 25 KHz at pin p2.3. Assume XTAL=22 MHZ. (6.5)

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Note: Attempt any five questions including Q no.1 which is compulsory.
Select one question from each unit. Assume suitable missing data if any.

Q1 Attempt all:

- (a) If the 8085 adds 87H and 79H specify the contents of the accumulator and the status of the S, Z and CY flags. (2)
- (b) How does the microprocessor differentiate among a positive number, a negative number and a bit pattern. (2)
- (c) What is the function of A1 and A0 lines in 8255? (2)
- (d) Explain the display RAM section of 8279? (2)
- (e) Explain the register banks of 8051? (2)
- (f) How Type 1 dedicated interrupt is generated. What is its vector location? (2)
- (g) How much time is taken by the 8085 microprocessor to execute CALL instruction if it is running at 2MHz? (2)
- (h) Explain the flag register of 8086 microprocessor. (4)
- (i) Explain the difference between MOV, MOV C and MOVX instruction in 8051. (4)
- (j) Check whether the following statements are true or false: (3)
 - (i) Instruction EI is necessary to implement the TRAP interrupt, but external hardware and the SIM instruction are unnecessary.
 - (ii) The execution of instruction MVI A, 10H and SIM will enable all three interrupts (RST 7.5, RST 6.5, and RST 5.5).
 - (iii) Instruction RIM is used to disable the interrupts RST 7.5, RST 6.5 and RST 5.5.

UNIT-I

- Q2 (a) Explain the function of following instructions with suitable examples: (6)
(i) DCX (ii) CPI (iii) XRA (iv) LDAX (v) EI (vi) CMC
(b) Draw and explain the timing diagram of MVI A, 32H instruction. (6.5)
- Q3 (a) Differentiate between the following: (6)
(i) CALL and RET
(ii) PUSH and POP
(b) Write an assembly language program to check a set of six signed numbers and add the positive number. The numbers are stored in memory locations starting from XX60H and the final result is expected to be less than FFH and stored in location XX70H. (6.5)

UNIT-II

- Q4 (a) Explain the functional block diagram of 8086 microprocessor? (6)
(b) Explain the interrupt structure of 8086 including details like exceptions, hardware and software interrupts and interrupt vector table. (6.5)
- Q5 (a) Explain the function of following assembler directives with suitable examples: (6)
(i) DB (ii) ASSUME (iii) EQU (iv) ORG (v) ENDP (vi) PTR

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- (b) Write a program to generate a delay of 1000 ms using an 8086 system that runs on 10 MHz frequency. (6.5)

UNIT-III

- Q6 (a) Explain the block diagram of 8255? (6)
(b) Design a programmable timer using 8254 and 8086. Interface at an address 0080H for counter 0 and write the following assembly language program. The 8086 and 8254 run at 6 MHz and 1.5 MHz respectively. (6.5)
(i) To generate a square wave of period 10 ms.
(ii) To interrupt the processor after 100 ms.
- Q7 (a) Explain the difference between 8253/8254. Briefly explain its different modes of operation? (6)
(b) Explain the functional block diagram of 8279? (6.5)

UNIT-IV

- Q8 (a) Explain the block diagram and flag register of 8051 microcontroller? (6)
(b) Explain the TMOD and TCON register? (6.5)
- Q9 (a) Describe interrupts used in 8051. Give their priority and address. (6)
(b) Write a program to generate a square wave of 5 KHz using timer 1 in mode 0 using 8051. (6.5)

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Maximum Marks: 75

Note: Attempt any five questions including Q.No1 which is compulsory.
Attempt one question from each unit. Assume suitable missing data if any.

- Q1 Attempt all: (2.5x10=25)
- (a) Explain the function of SIM instruction?
 - (b) Distinguish between ACALL AND LCALL instructions in 8051?
 - (c) What do you mean by maximum and minimum mode of 8086?
 - (d) Explain the function of the following signals:-
(i) EA (ii) NMI (iii) LOCK (iv) PSEN (v) TEST
 - (e) Explain the various addressing modes of 8086 microprocessors?
 - (f) Explain the data types and assembler directives of 8051?
 - (g) Explain the flag register of 8086?
 - (h) What is the use of BSR mode in 8255?
 - (i) Draw the timing diagram of instruction MVI B, 05H?
 - (j) Differentiate memory mapped I/O and I/O mapped I/O?

UNIT-I

- Q2 (a) Explain the various hardware and software interrupts of 8085 in detail and also calculate the vectored address of each interrupt? (6)
(b) Write an assembly language program to generate Fibonacci series along with the flow chart? (6.5)
- Q3 (a) Explain the function of following instructions with suitable examples:- (6)
(i) DAD (ii) DAA (iii) CMC (iv) JPO, JPE (v) EI, DI (vi) RET
(b) Explain the difference between partial and full address decoding scheme? Connect 16k bytes with the systems lines of microprocessor 8085 performing partial address decoding i.e use A₁₅ line for generating chip select logic. The memory IC available is EPROM with starting memory address is 9000H? (6.5)

UNIT-II

- Q4 (a) Explain the functional block diagram of 8086 microprocessor? (6)
(b) Write a delay program to create a time delay of 240ms. Assume that 5MHZ clock is used with 8086. Following is given:- (6.5)

INSTRUCTIONS	NO. OF T STATES
Mov BX, Immediate	4
DEC Register	2
NOP	8
JNZ LABEL	16
RET	8

- Q5 (a) Explain the functions of instructions with suitable examples:- (6)
(i) LEA (ii) AAA (iii) CWD (iv) MOVSB (v) REP (vi) LOOP
(b) What do you mean by segment override prefix? Write the instruction format for MOV DS: 2345 [BP], DX (6.5)

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UNIT-III

- Q6 (a) Explain the block diagram of 8251? (6)
(b) Explain the features of 8253. Briefly explain its different modes of operation? (6.5)
- Q7 (a) Write an ALP to generate square wave using 8255? (6)
(b) With the help of functional block diagram, explain the different features available in 8279? (6.5)

UNIT-IV

- Q8 (a) The following shows crystal frequency for three different 8051-based systems. Find the period of the machine cycle in each case (i) 15MHZ (ii) 30MHZ (iii) 40MHZ? (3)
(b) Explain the block diagram and PSW of 8051 microcontroller? (6)
(c) Explain the TMOD register? (3.5)
- Q9 (a) What is the difference between interrupts and polling? Explain various interrupts of 8051? (6)
(b) Write a program to generate two square waves one of 5KHZ frequency at pin P1.3 and another of frequency 25KHZ at pin P2.3. Assume XTAL=22MHZ?(6.5)
